BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 314 · Databases, SQL and web code practice

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does INNER JOIN require?

Q2. What is a foreign key?

Q3. How does label for connect to input?

Q4. Zero-indexed 4×3 array: total values and last index pair.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

join — Match rows across tables.

foreign key — Key relationship across tables.

label — Accessible form control description.

textContent — DOM text-setting property.

New learning

Understand — A primary key identifies rows; a foreign key creates the relationship across tables.

Apply — An INNER JOIN returns matching rows; a WHERE clause then filters them.

Evaluate — HTML structures user forms, CSS styles them, and JS can respond to events. Server-side checks remain authoritative.

Worked example

Write a form label correctly associated with email input.

1. Predict first — Matching for and id improves accessibility, while name defines the submitted field.

2. Explain and check — Explain the reason for the result.

We Do · Practise with guidance

Write JavaScript to set the paragraph id='status' to 'Saved'.

```
document.getElementById('status').textContent='Saved';
```

Predict first

textContent inserts safe text without parsing markup.

Explain and check

Explain the reason for the result.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Trace SQL JOIN and WHERE in the lesson example; identify final row and excluded rows. [6]

Question 2

Write SQL selecting names with marks≥70 using the two-table design. [6]

Question 3

Explain why a form's browser validation cannot be the sole protection for a school database, including one safe server procedure. [6]

Exit Ticket · Answer before checking

A client-side HTML required field makes server-side validation unnecessary.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Join condition matching rows.

Q2. Field referencing key in a related table.

Q3. It matches the input id attribute.

Q4. 12 values, last index (3,2).

You Do pages · Independent questions

Q1. ID1 joins but filtered because Mark60 [2]; ID2 unmatched excluded [2]; ID3 survives as Cleo80 [2].

Q2. SELECT Student.Name [1] FROM Student [1] INNER JOIN Score [1] ON Student.ID=Score.ID [2] WHERE Score.Mark>=70 [1].

Q3. Client bypass possible [2]; validate types/ranges server-side [2]; use parameterised SQL [1], authorisation [1].

Exit Ticket · Model answer

False. Clients can bypass checks.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: